

Understanding Variational Autoencoders from a mathematical view

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October 7, 2025

What I cannot create, I do not understand.

Outline

- 1 Introduction: The Problem of Generative Modeling
- 2 Variational Inference: The Foundation
- 3 Variational Autoencoders: Theory
- 4 VAE Variants and Extensions
- 5 Advanced Topics
- 6 Implementation and Practical Considerations
- 7 Recent Advances and Future Directions
- 8 Conclusion

The Challenge of Generative Modeling

Given a dataset $\mathcal{D} = \{x^{(i)}\}_{i=1}^N$ sampled from an unknown distribution $p^*(x)$, how can we:

- Learn a model $p_\theta(x)$ that approximates $p^*(x)$?
- Generate new samples from the learned distribution?
- Perform inference: $p_\theta(z|x)$ for latent variables z ?

Key challenges:

- ① High-dimensional data spaces
- ② Intractable likelihood computation
- ③ Latent variable inference
- ④ Scalable training procedures

Traditional Approaches and Their Limitations

- **Maximum Likelihood Estimation (MLE):**

$$\theta^* = \arg \max_{\theta} \sum_{i=1}^N \log p_{\theta}(x^{(i)}) \quad (1)$$

- **Problem:** Direct likelihood computation often intractable

- **Latent Variable Models:**

$$p_{\theta}(x) = \int p_{\theta}(x|z)p_{\theta}(z)dz \quad (2)$$

- **Problem:** Integration over latent space z is typically intractable

Solution: Variational inference to approximate intractable posteriors

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Variational Inference Framework

Given a latent variable model $p_\theta(x, z) = p_\theta(x|z)p_\theta(z)$, we want to:

- Approximate the intractable posterior $p_\theta(z|x)$
- Maximize the marginal likelihood $p_\theta(x)$

Key idea: Introduce a variational distribution $q_\phi(z|x)$ to approximate $p_\theta(z|x)$

$$\text{KL}(q_\phi(z|x) \| p_\theta(z|x)) = \mathbb{E}_{q_\phi(z|x)}[\log q_\phi(z|x)] - \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(z|x)] \quad (3)$$

Evidence Lower Bound (ELBO)

The KL divergence can be decomposed as:

$$\text{KL}(q_\phi(z|x) \| p_\theta(z|x)) = \log p_\theta(x) - \text{ELBO}(x; \theta, \phi) \quad (4)$$

where the **Evidence Lower Bound** is:

$$\text{ELBO}(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p_\theta(z)) \quad (5)$$

Key insight: Since $\text{KL} \geq 0$, we have:

$$\log p_\theta(x) \geq \text{ELBO}(x; \theta, \phi) \quad (6)$$

Maximizing ELBO $\Leftrightarrow$ Minimizing KL divergence $\Leftrightarrow$ Approximating posterior

The ELBO can be rewritten as:

$$\text{ELBO}(x; \theta, \phi) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p_{\theta}(z)) \quad (7)$$

Two components:

- ① **Reconstruction term:** $\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)]$
 - Encourages $q_{\phi}(z|x)$ to encode information about x
 - Maximizes likelihood of reconstructing x from z
- ② **Regularization term:** $-\text{KL}(q_{\phi}(z|x) \| p_{\theta}(z))$
 - Encourages $q_{\phi}(z|x)$ to be close to prior $p_{\theta}(z)$
 - Prevents overfitting to training data

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Encoder-Decoder Architecture

- **Encoder (Recognition model):** $q_{\phi}(z|x)$
 - Maps data x to latent representation z
 - Typically: $q_{\phi}(z|x) = \mathcal{N}(z; \mu_{\phi}(x), \sigma_{\phi}^2(x)I)$
- **Decoder (Generative model):** $p_{\theta}(x|z)$
 - Reconstructs data x from latent z
 - Typically: $p_{\theta}(x|z) = \mathcal{N}(x; \mu_{\theta}(z), \sigma_{\theta}^2 I)$ or Bernoulli
- **Prior:** $p_{\theta}(z) = \mathcal{N}(z; 0, I)$

The Reparameterization Trick

Problem: How to backpropagate through stochastic sampling $z \sim q_\phi(z|x)$?

Solution: Reparameterization trick

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (8)$$

Benefits:

- Gradient flows through deterministic path: $x \rightarrow \mu_\phi(x), \sigma_\phi(x) \rightarrow z$
- Stochasticity moved to ϵ (independent of parameters)
- Enables end-to-end training with backpropagation

Alternative approaches:

- Score function estimator (higher variance)
- Control variates

VAE Training Objective

For a dataset $\mathcal{D} = \{x^{(i)}\}_{i=1}^N$, the VAE loss is:

$$\mathcal{L}(\theta, \phi) = \sum_{i=1}^N \text{ELBO}(x^{(i)}; \theta, \phi) \quad (9)$$

Expanding the ELBO:

$$\mathcal{L}(\theta, \phi) = \sum_{i=1}^N \left[\mathbb{E}_{q_{\phi}(z|x^{(i)})} [\log p_{\theta}(x^{(i)}|z)] - \text{KL}(q_{\phi}(z|x^{(i)}) || p_{\theta}(z)) \right] \quad (10)$$

Training procedure:

- 1 Sample $x^{(i)} \sim \mathcal{D}$
- 2 Sample $\epsilon \sim \mathcal{N}(0, I)$
- 3 Compute $z^{(i)} = \mu_{\phi}(x^{(i)}) + \sigma_{\phi}(x^{(i)}) \odot \epsilon$
- 4 Compute reconstruction loss: $\log p_{\theta}(x^{(i)}|z^{(i)})$
- 5 Compute KL divergence: $\text{KL}(q_{\phi}(z|x^{(i)}) || p_{\theta}(z))$
- 6 Backpropagate gradients

Analytical KL Divergence

For Gaussian distributions, the KL divergence has a closed form:

$$\text{KL}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I)) = \frac{1}{2} \sum_{j=1}^J (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1) \quad (11)$$

where J is the dimension of the latent space.

Interpretation:

- μ_j^2 : Penalizes large mean values
- $\sigma_j^2 - \log \sigma_j^2 - 1$: Encourages $\sigma_j \approx 1$
- Total effect: Regularizes latent space to be close to standard Gaussian

Benefits:

- No sampling required for KL term
- Stable gradients
- Interpretable regularization

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β -VAE: Controlling the Regularization

Motivation: Balance between reconstruction and regularization

β -VAE objective:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - \beta \text{KL}(q_{\phi}(z|x) \| p_{\theta}(z)) \quad (12)$$

Effect of β :

- $\beta > 1$: Stronger regularization $\Rightarrow$ Better disentanglement
- $\beta < 1$: Weaker regularization $\Rightarrow$ Better reconstruction
- $\beta = 1$: Standard VAE

Disentanglement: β -VAE encourages learning of independent latent factors

VQ-VAE: Vector Quantized VAE

Key idea: Use discrete latent representations instead of continuous

Architecture:

- Encoder: $z_e = E(x)$ (continuous)
- Quantization: $z_q = \text{Quantize}(z_e)$ (discrete)
- Decoder: $\hat{x} = D(z_q)$

Quantization process:

$$z_q = \arg \min_k \|z_e - e_k\|^2 \quad (13)$$

where $\{e_k\}_{k=1}^K$ are learnable codebook vectors.

Benefits:

- Discrete representations (useful for language, etc.)
- Better compression
- Enables autoregressive modeling in latent space

Challenge: Quantization is non-differentiable

Solution: Straight-through estimator

$$\frac{\partial \mathcal{L}}{\partial z_e} = \frac{\partial \mathcal{L}}{\partial z_q} \quad (14)$$

Training objective:

$$\mathcal{L} = \log p(x|z_q) + \| \text{sg}(z_e) - e_k \|^2 + \beta \| z_e - \text{sg}(e_k) \|^2 \quad (15)$$

where $\text{sg}(\cdot)$ is stop-gradient operator.

Components:

- Reconstruction loss: $\log p(x|z_q)$
- Codebook loss: $\| \text{sg}(z_e) - e_k \|^2$ (updates codebook)
- Commitment loss: $\beta \| z_e - \text{sg}(e_k) \|^2$ (encourages encoder to commit)

WAE: Wasserstein Autoencoder

Motivation: Replace KL divergence with Wasserstein distance

WAE objective:

$$\mathcal{L} = \mathbb{E}_{p(x)} \mathbb{E}_{q_\phi(z|x)} [c(x, D(z))] + \lambda \mathcal{D}(q_\phi(z), p(z)) \quad (16)$$

where:

- $c(x, D(z))$: Reconstruction cost
- $\mathcal{D}(q_\phi(z), p(z))$: Distance between aggregated posterior and prior
- λ : Regularization weight

Benefits:

- More flexible than KL divergence
- Better mode coverage
- Can use different distance metrics

VAE-GAN: Combining VAE and GAN

Idea: Use GAN discriminator to improve VAE reconstruction quality

Architecture:

- VAE encoder-decoder: $x \rightarrow z \rightarrow \hat{x}$
- GAN discriminator: Distinguishes real vs reconstructed samples
- Generator: VAE decoder

Training objective:

$$\mathcal{L} = \mathcal{L}_{\text{VAE}} + \lambda \mathcal{L}_{\text{GAN}} \quad (17)$$

where:

- $\mathcal{L}_{\text{VAE}}$: Standard VAE loss
- $\mathcal{L}_{\text{GAN}}$: Adversarial loss for discriminator

Benefits:

- Better sample quality than VAE alone
- Maintains inference capabilities

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Hierarchical VAE

Motivation: Learn hierarchical latent representations

Architecture:

$$p_{\theta}(x, z_1, \dots, z_L) = p_{\theta}(x|z_1) \prod_{l=1}^{L-1} p_{\theta}(z_l|z_{l+1}) p_{\theta}(z_L) \quad (18)$$

Inference:

$$q_{\phi}(z_{1:L}|x) = \prod_{l=1}^L q_{\phi}(z_l|z_{<l}, x) \quad (19)$$

Benefits:

- Multi-scale representations
- Better modeling of complex data
- Hierarchical generation process

Conditional VAE (C-VAE)

Extension: Incorporate conditioning information

Model:

$$p_{\theta}(x, z|c) = p_{\theta}(x|z, c)p_{\theta}(z) \quad (20)$$

Inference:

$$q_{\phi}(z|x, c) = \mathcal{N}(z; \mu_{\phi}(x, c), \sigma_{\phi}^2(x, c)) \quad (21)$$

Applications:

- Conditional generation
- Controlled synthesis
- Multi-modal learning

Training:

$$\mathcal{L} = \mathbb{E}_{q_{\phi}(z|x, c)}[\log p_{\theta}(x|z, c)] - \text{KL}(q_{\phi}(z|x, c) \| p_{\theta}(z)) \quad (22)$$

Importance Weighted Autoencoder (IWAE)

Motivation: Better approximation of log-likelihood

IWAE objective:

$$\mathcal{L}_K = \mathbb{E}_{z_1, \dots, z_K \sim q_\phi(z|x)} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x, z_k)}{q_\phi(z_k|x)} \right] \quad (23)$$

Key properties:

- $\mathcal{L}_K \geq \mathcal{L}_{K-1}$ (monotonic improvement)
- $\lim_{K \rightarrow \infty} \mathcal{L}_K = \log p_\theta(x)$ (exact likelihood)
- $K = 1$ recovers standard VAE

Trade-offs:

- Better approximation with larger K
- Higher computational cost
- More complex gradients

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VAE Implementation Details

Network architectures:

- **Encoder:** $x \rightarrow \mu_\phi(x), \log \sigma_\phi^2(x)$
- **Decoder:** $z \rightarrow \mu_\theta(z)$ (for Gaussian) or logits (for Bernoulli)
- **Activation:** ReLU, ELU, or Swish
- **Normalization:** BatchNorm or LayerNorm

Training tricks:

- **KL annealing:** Gradually increase KL weight
- **Free bits:** Prevent posterior collapse
- **Warm-up:** Start with reconstruction only

Common Issues and Solutions

Posterior collapse:

- **Symptom:** $q_{\phi}(z|x) \approx p_{\theta}(z)$ (no information in latent)
- **Causes:** Strong decoder, weak encoder
- **Solutions:** KL annealing, free bits, stronger encoder

Blurry reconstructions:

- **Cause:** Gaussian decoder assumption
- **Solutions:** VAE-GAN, better decoder architectures

Mode collapse:

- **Symptom:** Limited diversity in generated samples
- **Solutions:** Better regularization, architectural improvements

Evaluation Metrics

Reconstruction quality:

- MSE, SSIM, FID (Fréchet Inception Distance)
- Perceptual losses

Generation quality:

- FID, IS (Inception Score)
- Human evaluation

Representation quality:

- Disentanglement metrics (β -VAE metric, MIG)
- Downstream task performance
- Interpolation smoothness

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Recent Advances

Transformer-based VAEs:

- VQ-VAE-2: Hierarchical discrete representations
- DALL-E: Text-to-image generation
- VQGAN: High-quality image generation

Improved training:

- NVAE: Normalizing flow decoder
- VDVAE: Very deep VAE
- VDM: Diffusion-based decoder

Applications:

- Text generation (GPT-style models)
- Image synthesis and editing
- Audio generation
- Scientific modeling

Open Questions and Future Directions

Theoretical understanding:

- Why do VAEs work despite approximation errors?
- Optimal choice of prior distributions
- Connection to other generative models

Technical challenges:

- Better posterior approximations
- Scalable training for large models
- Integration with other architectures

Applications:

- Multi-modal learning
- Scientific discovery
- Efficient compression
- Robust representations

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Summary

Key contributions of VAE:

- Scalable variational inference for latent variable models
- End-to-end training with reparameterization trick
- Foundation for many generative models

VAE variants address:

- Disentanglement (β -VAE)
- Discrete representations (VQ-VAE)
- Sample quality (VAE-GAN, WAE)
- Hierarchical modeling (HVAE)

Impact:

- Influenced modern generative models
- Enabled practical latent variable modeling
- Foundation for many applications

Questions

If you are interested in the questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. How does the choice of prior $p_\theta(z)$ affect VAE performance? What are the trade-offs between different priors?
2. Can you derive the gradient of the ELBO with respect to the encoder parameters ϕ ? How does the reparameterization trick make this tractable?
3. What are the theoretical guarantees of VAE? Under what conditions does the ELBO provide a good approximation to the true log-likelihood?
4. How would you extend VAE to handle sequential data (like in language modeling)? What are the key challenges?

Welcome inputs

Any comments?

Welcome your inputs!